



Beta PDF

$$\text{Beta}(\theta | a, b) = \frac{1}{B(a, b)} \cdot \theta^{a-1} (1-\theta)^{b-1}$$

- Slide 40 in the video 5 from week 1.

- after some algebraic manipulation, we arrive at the normalization constant $B(a, b) = \int_0^1 \theta^{a-1} (1-\theta)^{b-1} d\theta$

Example of unnormalized form of Beta distribution

$$p(\theta) \propto \theta^5 (1-\theta)^4 \quad \text{✓ we are missing the normalizing constant } B(a, b).$$

- we use $a-1=5 \Rightarrow a=6$

$b-1=4 \Rightarrow b=5$

$$\text{So, } p(\theta) \propto \theta^{6-1} (1-\theta)^{5-1} \Rightarrow p(\theta) = \frac{1}{B(6, 5)} \theta^5 (1-\theta)^4$$

Estimating proportions w/ Binomial distribution

- Binomial Distribution

Spse. we observe

- $N=7$ indep Bernoulli trials
 - $y_i = 1$ success, $y_i \in \{0,1\}$
 - $\theta \in [0,1]$ probability of success
- $\Rightarrow$ our dataset $D = \{N=7, y_i=1\}$

task 1.1 find the likelihood

$$\begin{aligned} p(y|\theta) &= p(y_i=1|N=7, \theta) = \binom{N}{y} \theta^y (1-\theta)^{N-y} \\ &= \binom{7}{1} \theta^1 (1-\theta)^{7-1} \\ &= \binom{7}{1} \theta (1-\theta)^6 \end{aligned}$$

exercise 1.2

- using the equation $\hat{\theta}_{MLE} = \arg \max_{\theta} p(y|N, \theta)$

$\rightarrow$ which for the Binomial Distribution is $\hat{\theta}_{MLE} = \frac{y}{N}$, where $N=7$ and $y=1$
 $= \frac{1}{7} = \underline{\underline{0.14}}$

95% confidence interval

$$\begin{aligned}\hat{\theta}_{MLE} \pm 1.96 \sqrt{\frac{\hat{\theta}_{MLE} (1 - \hat{\theta}_{MLE})}{N}} \\ = 0.14 \pm 1.96 \sqrt{\frac{0.14 (1 - 0.14)}{7}} = 0.29 \\ = 0.14 - 1.96 \sqrt{\frac{0.14 (1 - 0.14)}{7}} = -0.11\end{aligned}$$

Exercise 13

- What happens if we observe $y=0$, instead of $y=1$?

$$\begin{aligned}p(y=0|N=7, \theta) &= \binom{N}{y} \theta^y (1-\theta)^{N-y} \\ &= \binom{7}{0} 1 (1-\theta)^7\end{aligned}$$

Beta Prior

$$E[\theta] = \int_0^1 \theta p(\theta|a_0, b_0) d\theta = \frac{a_0}{a_0 + b_0}$$

$$V[\theta] = \int_0^1 (\theta - E[\theta])^2 p(\theta|a_0, b_0) d\theta = \frac{a_0 b_0}{(a_0 + b_0)^2 (a_0 + b_0 + 1)}$$

$$\theta_{mode} = \frac{a_0 - 1}{a_0 + b_0 - 2}$$

We assume Uniform Prior

Task 2.1: Compute prior mean and variance of $p(\theta)$, $a_0=1$, $b_0=1$

$$E[\theta] = \frac{a_0}{a_0 + b_0} = \frac{1}{1+1} = \underline{\underline{\frac{1}{2}}}$$

$$V[\theta] = \frac{a_0 b_0}{(a_0 + b_0)^2 (a_0 + b_0 + 1)} = \frac{1 \cdot 1}{(1+1)^2 (1+1+1)} = \underline{\underline{\frac{1}{12}}}$$

Beta Binomial Posterior Distribution $P(\theta|y)$

Task 2.2 Compute parameters a, b of the posterior

$$P(\theta|y) = \text{Beta}(\theta | a_0 + y, b_0 + N - y)$$

- we have $y = 1, N = 7, a_0 = 1, b_0 = 1$

$$\begin{aligned} P(\theta|y) &= \text{Beta}(\theta | \overset{a}{1+1}, \overset{b}{1+7-1}) \\ &= \text{Beta}(\theta | \overset{a}{2}, \overset{b}{7}) \end{aligned}$$

Task 2.3 Plot prior density, likelihood, and posterior density as a function of θ for $\theta \in [0, 1]$.

$$\begin{aligned} P(\theta|y, N) &= \frac{P(y|\theta, N) \cdot P(\theta)}{P(y)} \propto P(y|\theta, N) P(\theta) \\ &= P(y|\theta, N) \cdot P(\theta) \\ &= \underbrace{\binom{N}{y} \theta^y (1-\theta)^{N-y}}_{\text{Consider with } \theta} \cdot \underbrace{\text{Beta}(\theta | a_0=1, b_0=1)}_{\text{prior}} \\ &= \underbrace{\left(\binom{7}{1} \cdot \theta^1 (1-\theta)^{7-1} \right)}_{\text{Consider with } \theta} \cdot \underbrace{\left(\frac{1}{B(1,1)} \theta^{1-1} (1-\theta)^{1-1} \right)}_{\text{prior}} \\ &\propto \theta (1-\theta)^6 \cdot \underbrace{\frac{1}{1}}_1 \\ &= \underline{\underline{\theta (1-\theta)^6}} \end{aligned}$$

this matches the unnormalized Beta PDF, which has the form:

$$\text{Beta}(\theta | a_{\text{post}}, b_{\text{post}}) \propto \theta^{a_{\text{post}}-1} (1-\theta)^{b_{\text{post}}-1}$$

So we match the exponents:

$$\bullet \theta^1 = \theta^{a_{\text{post}}-1} \Rightarrow a_{\text{post}} = \underline{2}$$

$$\bullet (1-\theta)^6 = (1-\theta)^{b_{\text{post}}-1} \Rightarrow b_{\text{post}} = \underline{7}$$

Alternatively, we can check by conjugacy rule

$$a_{\text{post}} = a_0 + y = 1 + 1 = \underline{2}$$

$$b_{\text{post}} = b_0 + N - y = 1 + 7 - 1 = \underline{7}$$

Task 2.4 Compute the Map-estimator for Θ as well as the posterior mean

(hint: the MAP-estimator is the mode of the posterior density, i.e. $\Theta_{\text{MAP}} = \arg \max_{\Theta \in [0,1]} p(\Theta|y)$ and can be computed analytically for Beta-Binomial)

MAP $\Theta_{\text{mode}} = \frac{a_0 - 1}{a_0 + b_0 - 2}$ our posterior $P(\Theta|y) = \text{Beta}(2,7)$

$$\Theta_{\text{mode}} = \frac{a_0 - 1}{a_0 + b_0 - 2} = \frac{2 - 1}{2 + 7 - 2} = \frac{1}{7} \approx 0.143$$

Posterior mean $E[\Theta] = \int_0^1 \Theta p(\Theta|a_0, b_0) d\Theta = \frac{a_0}{a_0 + b_0}$ our posterior $P(\Theta|y) = \text{Beta}(2,7)$

$$E[P(\Theta|y)] = \frac{a_0}{a_0 + b_0}$$

$$= \frac{2}{2+7} = \frac{2}{9} \approx 0.222$$



task 2.5 Compute a 50%, 90% & 95% posterior credibility interval (CI)

• What is a CI?

• Given:

- a prior: belief about Θ as Beta dist. $p(\Theta) = \text{Beta}(\Theta|a,b)$
- a likelihood: from y successes in N Bernoulli trials (binomial): $p(y|\Theta) = \text{Bin}(y|N,\Theta)$
- posterior: $p(\Theta|y) \sim \text{Beta}(a+y, b_0+N-y)$

$$\Rightarrow \text{we want to find } \Theta_1, \Theta_2, \text{ s.t. } \int_{\Theta_1}^{\Theta_2} p(\Theta|D) d\Theta = 0.5$$

Intuition

Given the data, where does 50% of the posterior mass lie?

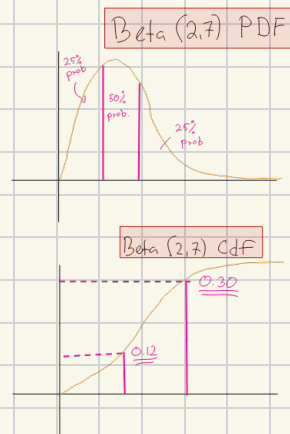
Math theory

posterior $\Theta|y \sim \text{Beta}(a_{\text{post}}, b_{\text{post}})$, where $a_{\text{post}} = a_0 + y$, $b_{\text{post}} = b_0 + N - y$

- we want to find the interval $[\Theta_1, \Theta_2]$ s.t. $\int_{\Theta_1}^{\Theta_2} \text{Beta}(\Theta|a_{\text{post}}, b_{\text{post}}) d\Theta = 0.5$

- we often choose a central interval, i.e. $\Theta_1 = F^{-1}(0.25)$, $\Theta_2 = F^{-1}(0.75)$

where F^{-1} is the inverse cumulative distribution function (CDF) of the Beta dist.



a) 50%

$$\int_{\theta_1}^{\theta_2} \text{Beta}(\theta|2,7) d\theta = 0.5 \Rightarrow$$

$$\theta_1 = \text{Beta}^{-1}(0.25; 2,7)$$

$$\theta_1 = 0.12$$

$$\theta_2 = \text{Beta}^{-1}(0.75; 2,7)$$

$$\theta_2 = 0.3$$

b) 90%

$$\int_{\theta_1}^{\theta_2} \text{Beta}(\theta|2,7) d\theta = 0.9 \Rightarrow$$

$$\theta_1 = \text{Beta}^{-1}(0.05; 2,7)$$

$$\theta_1 = 0.04$$

$$\theta_2 = \text{Beta}^{-1}(0.95; 2,7)$$

$$\theta_2 = 0.471$$

c) 95%

$$\int_{\theta_1}^{\theta_2} \text{Beta}(\theta|2,7) d\theta = 0.95 \Rightarrow$$

$$\theta_1 = \text{Beta}^{-1}(0.025; 2,7)$$

$$\theta_1 = 0.032$$

$$\theta_2 = \text{Beta}^{-1}(0.975; 2,7)$$

$$\theta_2 = 0.527$$

task 2.8 Show that posterior mean converges to MLE $\hat{\theta}_{MLE} = \frac{y}{N}$ as $N \rightarrow \infty$

• the posterior is $p(\theta|y) \sim \text{Beta}(a_0 + y, b_0 + N - y)$

• the mean is $E[\theta|y] = \frac{a_0 + y}{a_0 + b_0 + N}$

- using the hint of substituting $y = \hat{\theta}_{MLE} N$

$$= \frac{a_0 + \hat{\theta}_{MLE} \cdot N}{a_0 + b_0 + N}$$

- now taking the limit as $N \rightarrow \infty$ and divide each term by N

$$\lim_{N \rightarrow \infty} \frac{\frac{a_0}{N} + \hat{\theta}_{MLE}}{\frac{a_0}{N} + \frac{b_0}{N} + 1} = \underline{\underline{\hat{\theta}_{MLE}}}$$

What does this mean?

- in Bayes. inference, we combine prior belief & the data (likelihood)

-> as our dataset $N \uparrow$ the influence of prior diminishes and posterior is shaped almost entirely by the data

- so if we have little data, prior is very important

- as we get more data we agree w/ MLE

task 2.9 show that posterior mean is always between the prior mean $p(\theta) = \frac{a_0}{a_0+b_0}$ and $\hat{\theta}_{MLE}$, i.e. that posterior mean can be written as:

$$\mathbb{E}[\theta|D] = (1-\lambda)\theta_0 + \lambda \hat{\theta}_{MLE} \quad \forall \lambda \in [0, 1]$$

$$\begin{aligned} \mathbb{E}[\theta|D] &= \frac{a_0 + y}{a_0 + b_0 + N} \\ &= \frac{a_0}{a_0 + b_0 + N} + \frac{y}{a_0 + b_0 + N} \\ &= \frac{\theta_0}{\theta_0} \frac{a_0}{a_0 + b_0 + N} + \frac{\hat{\theta}_{MLE}}{\hat{\theta}_{MLE}} \frac{y}{a_0 + b_0 + N} \\ &= \theta_0 \frac{a_0}{a_0 + b_0 + N} + \hat{\theta}_{MLE} \frac{y}{a_0 + b_0 + N} \\ &= \theta_0 \frac{a_0 + b_0}{a_0 + b_0 + N} + \hat{\theta}_{MLE} \frac{N}{a_0 + b_0 + N} \\ &\text{- now define } \lambda = \frac{N}{a_0 + b_0 + N} = 1 - \lambda = \frac{a_0 + b_0}{a_0 + b_0 + N} \\ \therefore \mathbb{E}[\theta|D] &= (1-\lambda)\theta_0 + \lambda \hat{\theta}_{MLE} \end{aligned}$$

Implications:

- posterior mean is ALWAYS between the prior mean and the MLE

$$\mathbb{E}[\theta|D] \in [\min(\theta_0, \hat{\theta}_{MLE}), \max(\theta_0, \hat{\theta}_{MLE})]$$

- the posterior estimate Never goes outside what either the prior knowledge or the data is

- the weight on the MLE is

$$\lambda = \frac{N}{a_0 + b_0 + N}$$

- when $N \downarrow \rightarrow$ prior dominates

- when $N \uparrow \rightarrow$ the likelihood dominates

Task 3.1 Identify parameters a, b of the pdf, $p_a(\theta) = \frac{1}{Z_d} \theta^{36} (1-\theta)^{41}$, where $Z_d > 0$.

- we notice that the pdf looks like the functional form of the Beta distribution

Beta PDF

$$\text{Beta}(\theta|a,b) = \frac{1}{B(a,b)} \cdot \theta^{a-1} (1-\theta)^{b-1}$$

$$p(\theta|a,b) = \frac{1}{Z_d} \theta^{\overbrace{36}^a} (1-\theta)^{\overbrace{41}^b}$$

$$a-1=36 \Rightarrow a=37$$

$$b-1=41 \Rightarrow b=42$$

$$\text{Beta}(37, 42)$$

Task 3.2 Compute Z_d

$$\int p_a(\theta) d\theta = 1 \Rightarrow 1 = \int p_a(\theta) d\theta = \int \frac{1}{Z_d} \theta^{36} (1-\theta)^{41} d\theta = \frac{1}{Z_d} \int \theta^{36} (1-\theta)^{41} d\theta = \frac{1}{Z_d} \int \theta^{36-1} (1-\theta)^{42-1} d\theta$$

$$\Rightarrow Z_d = \text{Beta}(37, 42) \approx 1.099 \cdot 10^{-24}$$

Task 33 Compute the analytical expression for the model evidence for Beta-Binomial model

$$p(y) = \int \underbrace{p(y, \theta)}_{\text{Joint prob.}} d\theta$$

$$= \int \underbrace{p(y|\theta)}_{\text{Product rule}} p(\theta) d\theta$$

$$= \binom{N}{y} \theta^y (1-\theta)^{N-y} \cdot \frac{1}{\text{Beta}(a, b)} \theta^{a-1} (1-\theta)^{b-1} d\theta$$

$$= \binom{N}{y} \frac{1}{\text{Beta}(a, b)} \int \theta^y (1-\theta)^{N-y} \theta^{a-1} (1-\theta)^{b-1} d\theta$$

$$= \binom{N}{y} \frac{1}{\text{Beta}(a, b)} \int \theta^{y+a-1} (1-\theta)^{N-y+b-1} d\theta$$

$$= \binom{N}{y} \frac{1}{\text{Beta}(a, b)} \int \theta^{y+a-1} (1-\theta)^{N-y+b-1} d\theta$$

$$= \binom{N}{y} \frac{1}{\text{Beta}(a, b)} \text{Beta}(y+a, N-y+b)$$

$$= \binom{N}{y} \frac{\text{Beta}(y+a, N-y+b)}{\text{Beta}(a, b)}$$

